such internal lines can be assigned an independent four-momentum. We can think of the independent momentum variables as characterizing the momenta that circulate in each loop. In particular a *tree* graph is one without loops; after taking the delta functions into account there are no momentum-space integrals left for such graphs.

For instance, in a theory with interaction (6.1.18), the S-matrix (6.1.27) for fermion–boson scattering is given by the momentum-space Feynman rules as

$$\begin{aligned}
&S_{\mathbf{p}_1{}'\sigma_1'n_1'\,\mathbf{p}_2{}'\sigma_2'n_2',\;\mathbf{p}_1\sigma_1n_1\,\mathbf{p}_2\sigma_2n_2} = \\
&\sum_{k'l'm'klm}(-i)^2(2\pi)^8 g_{l'm'k'}\,g_{mlk}\,u_{l'}^*(\mathbf{p}_1{}'\sigma_1'n_1')\,u_l(\mathbf{p}_1\sigma_1n_1) \\
&\times\int d^4q\,\left(-i(2\pi)^{-4}\frac{P_{m'm}(q)}{q^2+m_m^2-i\epsilon}\right) \\
&\times(2\pi)^{-6}\Big[u_{k'}^*(\mathbf{p}_2{}'\sigma_2'n_2')u_k(\mathbf{p}_2\sigma_2n_2)\delta^4(p_2-p_{2'}-q)\delta^4(p_1+q-p_{1'}) \\
&\quad+\;u_k^*(\mathbf{p}_2{}'\sigma_2'n_2')u_{k'}(\mathbf{p}_2\sigma_2n_2)\delta^4(p_2-p_{1'}-q)\delta^4(p_1-p_{2'}+q)\Big]\,,
\end{aligned}$$

with labels 1 and 2 here denoting fermions and bosons, respectively. The momentum-space integral here is trivial, and gives

$$\begin{aligned}
&S_{\mathbf{p}_1{}'\sigma_1'n_1'\,\mathbf{p}_2{}'\sigma_2'n_2',\;\mathbf{p}_1\sigma_1n_1\,\mathbf{p}_2\sigma_2n_2} = i(2\pi)^{-2}\delta^4(p_1+p_2-p_1'-p_2') \\
&\times\sum_{k'l'm'klm} g_{l'm'k'}\,g_{mlk}\,u_{l'}^*(\mathbf{p}_1{}'\sigma_1'n_1')\,u_l(\mathbf{p}_1\sigma_1n_1) \\
&\times\left[\frac{P_{m'm}(p_{1'}-p_1)}{(p_{1'}-p_1)^2+m_m^2-i\epsilon}\;u_{k'}^*(\mathbf{p}_2{}'\sigma_2'n_2')u_k(\mathbf{p}_2\sigma_2n_2)\right. \\
&\quad+\left.\frac{P_{m'm}(p_{2'}-p_1)}{(p_{2'}-p_1)^2+m_m^2-i\epsilon}\;u_k^*(\mathbf{p}_2{}'\sigma_2'n_2')u_{k'}(\mathbf{p}_2\sigma_2n_2)\right]\,.
\end{aligned}\tag{6.3.5}$$

In the same way, the S-matrix element (6.1.28) for fermion–fermion scattering in the same theory is

$$\begin{aligned}
&S_{\mathbf{p}_1{}'\sigma_1'n_1'\,\mathbf{p}_2{}'\sigma_2'n_2',\;\mathbf{p}_1\sigma_1n_1\,\mathbf{p}_2\sigma_2n_2} = i(2\pi)^{-2}\delta^4(p_1+p_2-p_1'-p_2') \\
&\times\sum_{k'l'm'klm} g_{m'mk'}\,g_{l'lk}\frac{P_{k'k}(p_{1'}-p_1)}{(p_{1'}-p_1)^2+m_k^2-i\epsilon} \\
&\times u_{m'}^*(\mathbf{p}_2{}'\,\sigma_2'\,n_2')\,u_{l'}^*(\mathbf{p}_1{}'\,\sigma_1'\,n_1')\,u_m^*(\mathbf{p}_2\,\sigma_2\,n_2)u_l^*(\mathbf{p}_1\,\sigma_1\,n_1) \\
&-[1'\rightleftharpoons 2']\,.
\end{aligned}\tag{6.3.6}$$

These results illustrate the need for a more compact notation. We may define a fermion–boson coupling matrix

$$[\Gamma_k]_{lm}\equiv g_{lmk}\,.\tag{6.3.7}$$

The matrix elements (6.3.5) and (6.3.6) for fermion–boson and fermion–